



AGENT ORCHESTRATION

What breaks past 10K invocations per day.

Swipe →





THE SCALE CLIFF



500/day works. 10K/day exposes **failure modes.**

- State consistency degrades under concurrent session load
- Memory layer throughput becomes the bottleneck
- Retry storms from cascading failures amplify errors
- Tool call latency variance widens with agent count





FAILURE MODE 1: STATE



State consistency breaks at scale.

- Shared memory stores hit lock contention above 1K concurrent
- Session state written to wrong context under race conditions
- Long-running agent tasks lose context on checkpoint gaps
- State recovery logic not tested under concurrent workload





FAILURE MODE 2: MEMORY



Memory layer throughput collapses.

- Vector search latency spikes at high query volume
- Embedding generation becomes blocking at scale
- Memory consolidation jobs interfere with live inference
- Cache invalidation logic not designed for multi-agent writes





FAILURE MODE 3: TOOL CALLS



External tool calls amplify failures.

- Rate limits on external APIs hit at orchestration scale
- Retry logic creates cascading load on downstream services
- Tool call timeout logic not consistent across agent types
- Error propagation from tools not handled gracefully





FAILURE MODE 4: OBSERVABILITY



You can't debug what you can't trace.

- Agent decision traces missing at high invocation rate
- Log sampling drops critical failure context
- Metric granularity too coarse to identify root cause
- No tracing correlation across multi-agent workflows





ENGINEERING SOLUTIONS



What the 10K/day stack looks like.

- Distributed state management with optimistic locking
- Async memory operations decoupled from inference path
- Tool call circuit breakers with graceful degradation
- Distributed tracing correlated across all agent hops





MONITORING AT SCALE



Metrics that catch failure early.

- P95/P99 tool call latency tracked per tool per agent type
- State write failure rate as leading indicator
- Memory layer queue depth as capacity signal
- Escalation rate spike as orchestration health proxy





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